Alafia(PRD)

📌 Proje Over
🎯 Obje
Probl State
C
C
P
P
Goals and Succ Metri
P
G
M
S
M
U
P
S
M
/
K
K
F

📌 Project Overview

Pro Ala	**Pre By: Pro**	**Date Created: 3rd March, 2026**
Las Up 3rc Ma 20	**Ver Ver 1**	

🎯 Objective

Alafia is a cloud-based, patient-owned EMR platform built on Web5 principles. It gives patients in Nigeria and West Africa full sovereignty over their health data through a decentralised identity (DID) wallet, while enabling hospitals, clinics, pharmacies, and diagnostic labs to collaborate on a shared, permissioned network.

Alafia solves three interconnected problems:

- **Fragmented records** — data siloed across hospitals in paper files and non-interoperable systems, causing repeated tests, delays, and

harmful errors.

- **No patient data ownership** — patients have zero visibility or control over who accesses their records or for how long.
- **Broken last-mile care delivery** — after a consultation, patients must physically navigate pharmacies and labs with no integrated support.

Think of it as a digital folder for every patient, available whenever and wherever they need care whether it's at a city hospital or a small clinic in a rural area.

Problem Statement

Context

Nigeria's healthcare system has low EMR adoption. Most records are paper files tied to a single facility. When patients move between hospitals — by choice, referral, or emergency — their history does not travel with them.

Core Pain Points

Stakeholder	Pain Point
Patient	No ownership of records. Must repeat tests at every hospital. No visibility into who has accessed their data.
Doctor / Nurse	Incomplete patient history at point of care. Manual, error-prone entry. No instant sync.
Lab Technician	Results printed or emailed separately; no integration with the patient's record.
Hospital Admin	No audit trail. High admin overhead. Compliance risk.
Pharmacist	Cannot verify prescriptions digitally. No link between EMR and dispensing.

P
—
La
In
📐 U
&
D
🧬 T
R
🧱 D
🗓️ T
🧠 O
Q
/
R
R
PDLC
and
PLC
Positi
Open
Ques
Gloss
📚 S
D

Goals and Success Metrics

Product Goals

- Give patients full ownership and portability of health records via a Web5 DID wallet.
- Enable real-time, cross-hospital record sync for healthcare professionals.
- Connect patients with pharmacies and labs in a seamless post-consultation workflow.
- Build a sustainable B2B2C business with transaction-based and subscription revenue.

MVP Success Metrics

Metric	Target
Patient self-onboarding	Functional end-to-end
Hospital-invited onboarding	Functional end-to-end
Real-time data entry & sync	< 5 seconds cross-device
DID wallet — grant / revoke	Functional with audit log
Pharmacy order + payment	Functional via Interswitch
Lab drop-off + digital result	Functional with notification
Record deletion from hospital DB	Patient-initiated, irreversible

👤 User Personas

> *Who are you building this for?*

- Primary audience — Hospitals
- Secondary audience (if any) — Patients
- Alafia User Personas

✅ Success Metrics / KPIs

> *How will we measure success?*

- At least 50 hospitals onboarded and actively using the app within the first year, with 90% reporting seamless integration.
- Maintain over 98% accuracy of stored medical records and over 95% completeness of required patient data fields.
- Ensure average record retrieval time < 3 seconds and system uptime of 99.9%.
- Achieve 100% compliance with International and Local Health Boards and zero data breaches annually.
- Reach a minimum 85% satisfaction score in patient and hospital feedback surveys

🧩 Key Features / Requirements

> *What exactly needs to be built?*

Patient DID Wallet (Web5)

- DID created on first registration (self-sovereign).
- Records stored as encrypted Verifiable Credentials (VCs) in patient's PDS.
- Scoped access tokens issued per hospital, per episode of care.
- Three revocation modes: manual, time-limit expiry, auto-expire on discharge.
- Audit log visible to patient only.

Real-Time Data Entry

- Fields: vitals (BP, temp, HR, SpO2, weight), diagnoses (ICD-10), prescriptions, progress notes, allergies, surgical history.
- Auto-save every 10 seconds.
- WebSocket sync, < 5s latency target.
- Offline: IndexedDB / SQLite cache, background sync queue.

Hospital-to-Hospital Transfer

- Patient-initiated from their wallet; or hospital-requested with patient approval.
- Standalone mode (Alafia as the EMR) or hybrid (HL7 FHIR sync with existing systems).
- Emergency override: one-click share with audit log entry.

Pharmacy Integration

- Auto-match on prescription creation by proximity + stock.
- Interswitch checkout (card, bank transfer, USSD).
- Digital prescription sent to pharmacy with patient consent.
- Push + SMS delivery tracking.

Lab Order & Result Delivery

- Auto-match to nearest partner lab on lab order creation.
- Patient receives address, reference number, and preparation instructions.
- Patient drops off sample physically.
- Lab uploads results → pushed to patient wallet + doctor dashboard.
- Patient controls sharing and revocation of results.

Role-Based Access Control

Role	View Records	Edit Records	Admin
Patient	Own only	Via wallet	Manage grants
Doctor	Full (if granted)	Full clinical entry	—
Nurse	Assigned patients	Vitals only	—
Lab Tech	Lab orders only	Results upload	—
Admin	Metadata only	—	User mgmt, billing

🖥️ User Stories

Patient — Data Ownership & Wallet

- As a patient, I want a DID-based health wallet so I own my records independently of any hospital.
- As a patient, I want to grant hospitals time-limited access to my records.
- As a patient, I want to revoke a hospital's access (including on discharge) so my records are deleted from their database.
- As a patient, I want to set an access expiry date so permissions lapse automatically.

Acceptance Criteria:

- DID wallet created on sign-up.
- Access grant generates a permissioned token with configurable expiry.
- Revocation deletes hospital's local copy immediately; patient's PDS retains the canonical record.
- Audit log records every grant, view, edit, and revocation with timestamp and actor.

Healthcare Professional — Real-Time Data Entry

- As a doctor, I want instant access to a patient's complete history for better treatment decisions.
- As a nurse, I want to update vitals in real-time so all caregivers see the most recent readings.
- As a lab tech, I want results to sync immediately so the doctor can review without delay.

Acceptance Criteria:

- Data entered in Hospital A is visible in Hospital B within 5 seconds.
- Offline entry queues and syncs on reconnection.
- Sync status indicator always visible.
- RBAC enforced: doctors (read/write), nurses (vitals), lab techs (results upload), admins (config).

Patient — Pharmacy Integration

- As a patient, I want to be auto-matched with nearby pharmacies that stock my prescription.
- As a patient, I want to pay in-app and have medication delivered.

Acceptance Criteria:

- System auto-matches by proximity and stock on prescription creation.
- Payment via Interswitch (card, bank transfer, USSD).
- Pharmacy receives verified digital prescription.
- Patient gets real-time delivery notifications.

Patient — Lab Integration

- As a patient, I want to be matched with a nearby lab for sample drop-off.
- As a patient, I want digital results sent to my wallet and shared with my doctor.

Acceptance Criteria:

- Lab order auto-matches to nearest partner lab.
- Patient receives: lab address, reference number, preparation instructions.
- Results pushed to patient wallet and doctor dashboard simultaneously.
- Patient controls access to results at all times.

📐 UX & Design

> *Include any design constraints or brand guidelines.*

- Prototype: [Link to Figma or Notion page]
- UI mockups: [Add image/file or embed Figma]
- Design notes:

🛠️ Technical Requirements

Layer	Technology	Notes
Frontend	React Native + React (web)	Cross-platform
Backend	Node.js / NestJS	REST + WebSocket
Database	PostgreSQL + Redis	Relational + cache
Web5 / DID	TBD Web5 SDK / Veramo	Decentralised identity
Cloud	AWS / GCP Africa region	Low-latency
Payment	Interswitch Quickteller API	Card, transfer, USSD
EMR Integration	HL7 FHIR R4	Hybrid mode
Security	AES-256 at rest, TLS 1.3	NDPR + HIPAA-aligned

Data Architecture — Web5 Model: The patient's PDS is the single source of truth. MEDrec's cloud holds only encrypted routing metadata and access control records — never raw health data.

> Revocation deletes the hospital's copy, invalidates the token, and writes immutably to the audit log.

🧱 Dependencies

> *What needs to happen before this can be built?*

- [ ] Interswitch API sandbox credentials.
- [ ] Web5 DID SDK.
- [ ] Partner pharmacies and labs before beta.
- [ ] Hospital cooperation for FHIR hybrid mode.

🗓️ Timeline

> *Key dates and phases*

Alafia MVP Timeline

🧠 Open Questions / Risks

Risks

Risk	Likelihood	Mitigation
Cybersecurity breach	Medium	E2E encryption, pen tests, no central data store
Low hospital adoption	High	B2B2C — patients drive demand
Poor connectivity	High	Offline-first with sync queue
Staff resistance	Medium	Simple UI, training, phased rollout
Web5 DID immaturity	Medium	OAuth2 fallback while DID matures
Pharmacy/lab no-shows	Medium	SLAs + patient rating system

PDLC and PLC Positioning

PDLC: Currently in Design & Development. The 2-week MVP delivers the first testable prototype. Post-MVP enters Testing & Iteration with live hospital pilots.

PLC: Alaenters at the Growth phase of Nigeria's digital health curve. The Web5 data ownership layer is a meaningful differentiator — no incumbent Nigerian EMR currently offers patient-controlled records.

Open Questions

- Which DID library for MVP — TBD Web5 SDK, Veramo, or Spruce DIDKit?
- Web-only MVP or web + mobile simultaneously?
- Minimum pharmacy/lab partners needed before public beta?
- What clinical conditions permit emergency record access override, and who governs this?
- Is the hospital admin dashboard in or out of MVP scope?

Glossary

Term	Definition
DID	Decentralised Identifier — globally unique ID controlled by the patient.
PDS	Personal Data Store — patient's encrypted data vault.
VC	Verifiable Credential — tamper-proof digital health record attached to the patient's DID.
FHIR	Fast Healthcare Interoperability Resources — international health data exchange standard.
NDPR	Nigeria Data Protection Regulation.
RBAC	Role-Based Access Control.
B2B2C	Business-to-Business-to-Consumer — Alafia sells to hospitals, who serve patients.

Term	Definition
Web5	Data sovereignty architecture: Web2 usability + decentralised identity + data ownership.
Interswitch	Nigeria's leading payment infrastructure (Quickteller / WebPay).*MEDrec — Putting patients in control of their health stor*

📚 Supporting Documents

- Research: [Link]
- Competitive Analysis: [Link]
- Previous version(s): [Link]

Supporting Documents

Document Name	Document Type	Document Link
User Journey Maps	Design	
Technical Architecture Specifications	Technical Documentation	
Market Research Analysis Report	Market Analysis	